

Len 20

Recap

$$\textcircled{1} E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} P(X > x) dx - \int_0^{\infty} P(X < -x) dx$$

$$\textcircled{2} E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Discrete R.V.s

$$E[g(X)] = \sum_{x \in X_0} g(x) P_X(x)$$

$$\textcircled{3} \text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

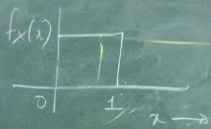
$$\begin{aligned} \textcircled{4} E[aX + b] &= \int_{-\infty}^{\infty} (ax + b) f_X(x) dx \\ &= a \int_{-\infty}^{\infty} x f_X(x) dx + b \int_{-\infty}^{\infty} f_X(x) dx \\ &= a E[X] + b \end{aligned}$$

$$\textcircled{5} \text{Var}(aX + b) = a^2 \text{Var}(X)$$

Examples:

(a) $X \sim \text{Uniform}[0, 1]$

$$f_X(x) = \begin{cases} 1 & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$



$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^1 x \cdot 1 dx = \left. \frac{x^2}{2} \right|_0^1 = \frac{1}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^1 x^2 \cdot 1 dx = \left. \frac{x^3}{3} \right|_0^1 = \frac{1}{3}$$

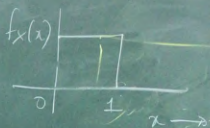
$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$= \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}$$

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$$X \sim \text{Uniform}(\{1, 2, \dots, n\}).$$

$$P_X(x) = \begin{cases} c & x \in \{1, 2, \dots, n\} \\ 0 & \text{otherwise} \end{cases}$$

$$c = \frac{1}{n} \quad \Leftarrow \quad \sum_{x=1}^n P_X(x) = 1$$

$$\begin{aligned} E[X] &= \sum_{x=1}^n x P_X(x) \\ &= \sum_{x=1}^n x \times \frac{1}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2} \end{aligned}$$

$$E[X^2] = \sum_{x=1}^n x^2 P_X(x)$$

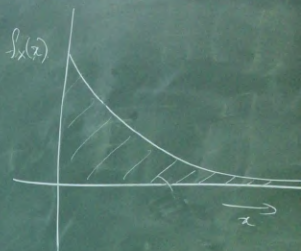
$$= \frac{1}{n} \frac{n(n+1)(2n+1)}{6}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

⑥

$X \sim \text{Exponential}(\lambda)$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$\begin{aligned} \int_{-\infty}^{\infty} f_X(x) dx &= \int_0^{\infty} \lambda e^{-\lambda x} dx \\ &= \lambda \left(\frac{e^{-\lambda x}}{-\lambda} \right) \Big|_0^{\infty} \\ &= - (0 - 1) = 1 \end{aligned}$$

Memoryless Property

$$P(X > t + \tau | X > \tau) = P(X > t).$$

$$P(X > t) = 1 - P(X \leq t)$$

$$= 1 - F_X(t)$$

$$= 1 - \int_0^t \lambda e^{-\lambda x} dx$$

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

↓

$$= 1 - \left[\lambda \left(\frac{e^{-\lambda x}}{-\lambda} \right) \right]_0^t$$

$$= 1 - [e^{-\lambda t} - 1]$$

$$= e^{-\lambda t}$$

$$P(X > t + \tau | X > \tau) = \frac{P(\{X > t + \tau\} \cap \{X > \tau\})}{P(\{X > \tau\})} = \frac{P(X > t + \tau)}{P(X > \tau)}$$

$$= \frac{e^{-(t+t)\lambda}}{e^{-t\lambda}} = e^{-t\lambda}$$

$$= P(X > t)$$

Exponential RV Satisfies
Memoryless property

Incoming:



time of arrival
between two packets
is considered as
exponential RV (λ_1).

outgoing
served



time taken to
process each
packet is
modeled as
exponential
RV (λ_2).

$$\lambda_2 > \lambda_1$$

$$E[X] = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

$$E[X] = \int_0^{\infty} \underbrace{x}_{u(x)} \underbrace{e^{-\lambda x} \lambda}_{v'(x)} dx$$

$$= \left[x \left(\frac{e^{-\lambda x} \lambda}{-\lambda} \right) \right]_0^{\infty}$$

$$- \int_0^{\infty} 1 \cdot \left(\frac{e^{-\lambda x} \lambda}{-\lambda} \right) dx$$

$$= -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$- \left[\lim_{x \rightarrow \infty} x e^{-\lambda x} - 0 x e^{-\lambda \cdot 0} \right] + \frac{1}{\lambda}$$

$$\lim_{x \rightarrow \infty} x e^{-\lambda x} = 0 \Rightarrow E[X] = \frac{1}{\lambda}$$

$\frac{x}{e^{\lambda x}}$

$$\int_a^b u(t) v'(t) dt$$

$$= u(t) v(t) \Big|_a^b - \int_a^b u'(t) v(t) dt$$

$$\frac{d}{dt} u(t) v(t) = u'(t) v(t) + u(t) v'(t)$$

$$\lim_{t \rightarrow \infty}$$

$$\begin{aligned}
 E[X^2] &= \int_0^{\infty} \underbrace{x^2}_{u(x)} \underbrace{e^{-\lambda x} \lambda}_{v'(x)} dx \\
 &= -x^2 e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx \\
 &= 0 + 2 \frac{E[X]}{\lambda} = \frac{2}{\lambda^2}
 \end{aligned}$$

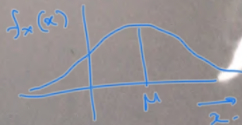
$$\lim_{x \rightarrow \infty} x^2 e^{-\lambda x} = 0$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - (E[X])^2 \\
 &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}
 \end{aligned}$$

Normal RV (Gaussian RV)

$$X \sim N(\mu, \sigma^2)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{x-\mu}{\sigma}\right)^2 \frac{1}{2}}$$



$$x \in \mathbb{R}$$

Motivation to study Normal R.Vs

$$X_1, X_2, X_3, \dots, X_n$$

identical & independent (i.i.d).

$$\frac{X_1 + \dots + X_n}{n}$$

$X_1, X_2, \dots, X_n$ are independent.

$$F_{X_i}(x) = F_{X_j}(x) \quad \forall i$$

The averages follow normal distribution

$$\boxed{X_{11}, \dots, X_{n1}}$$

are iid $\text{Ber}(p)$

$$X_{12}, \dots, X_{n2}$$

$\dots \text{Ber}(p)$

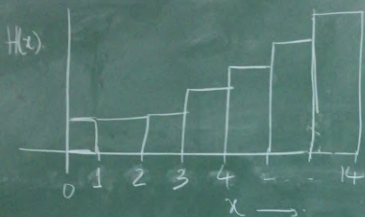
$$\vdots$$

$$X_{1M}, \dots, X_{nM}$$

$$\dots X_{nM}$$

$$S_i = \frac{X_{1i} + X_{2i} + \dots + X_{ni}}{n}$$

Histogram example



1/4 marks.

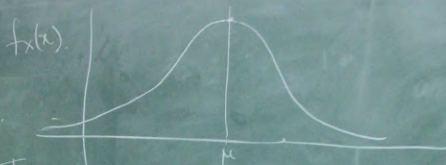
$H(x)$ = # of people with marks

in the interval $[x], [x+1)$ if x is not integer
 $[x, x+1)$

① Plot histogram $\bar{S}_i$
average



looks like pdf of normal distribution



To check

$$\textcircled{1} \int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\textcircled{2} E[X] = \mu$$

$$\textcircled{3} \text{Var}(X) = \sigma^2$$

not integer

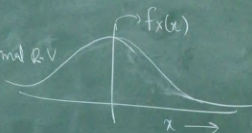
$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$dy = \frac{dx}{\sigma} \quad y = \frac{x-\mu}{\sigma} = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

If $X \sim N(0, 1)$ it is referred to as standard normal r.v.

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad x \in \mathbb{R}.$$

pdf of
Standard normal r.v.



$$\text{Let } I = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy$$

$$\begin{aligned} I^2 &= \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \right] \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \right] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} dx dy \end{aligned}$$

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}} dx dy.$$

$$= \int_0^{2\pi} \int_0^{\infty} \frac{1}{2\pi} e^{-r^2/2} r dr d\theta.$$

$$= 2\pi \times \frac{1}{2\pi} \int_0^{\infty} e^{-r^2/2} r dr$$

$$= \int_0^{\infty} e^{-t} dt$$

$$= 1$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$J(r, \theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta & \sin \theta \\ -r \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r$$

$$r^2 = t$$

$$r dr = dt$$

$$\Rightarrow \left(\int_{-\infty}^{\infty} f_x(x) dx \right)^2 = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} f_x(x) dx = 1$$